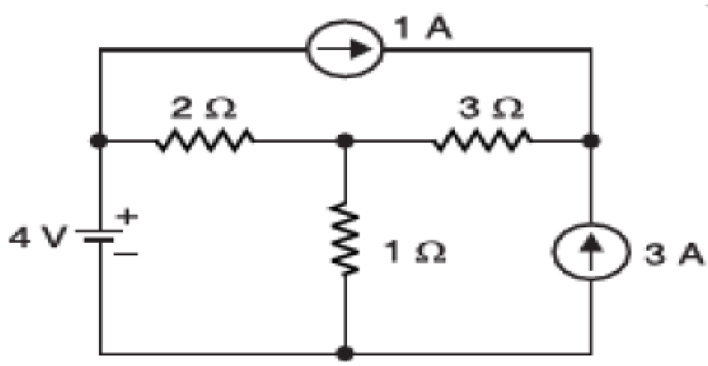
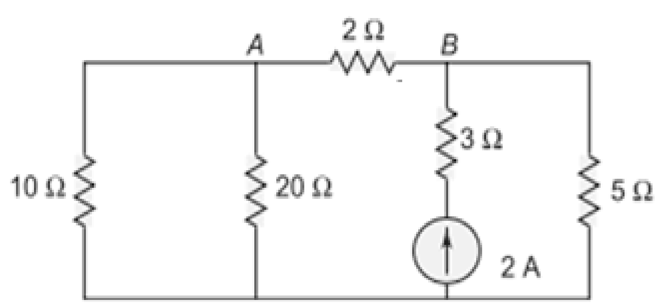
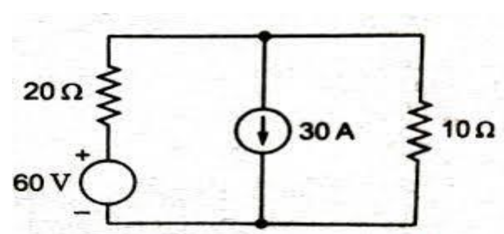
ABES Engineering College, Ghaziabad

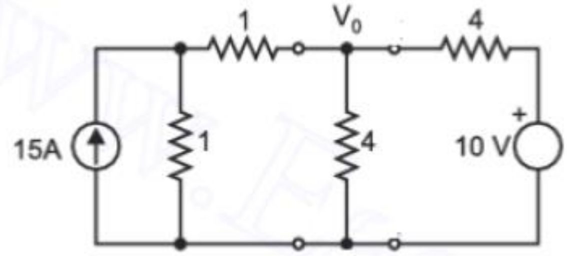
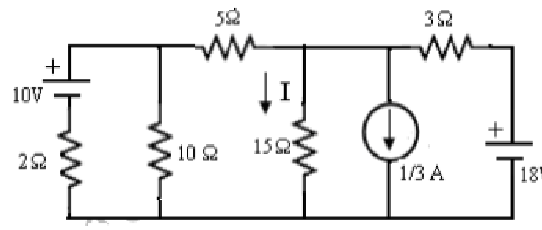
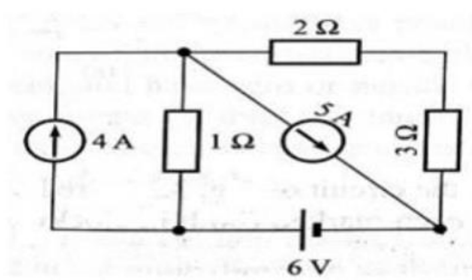
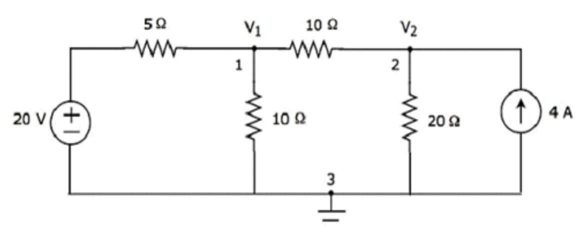
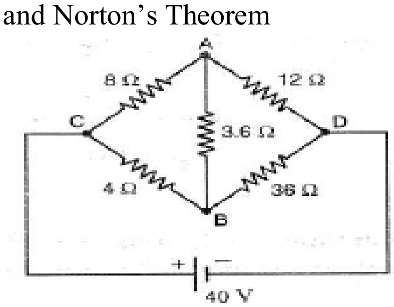
Department of Applied sciences & Humanities

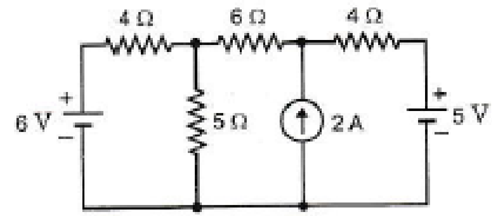
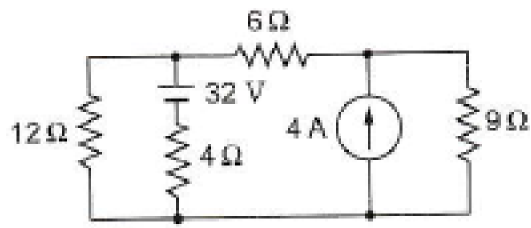
Session: 2025-26 **Semester:** II **Section:**

Course Code: 25EL201 **Course Name:** Basics of Electrical Engineering

<u>Assignment Sheet-1 (Unit-1)</u>

Q. No.	KL	CO	Question	Ans.
Q-1	**K3**	**CO1**	. Using Mesh analysis, find current in 1 Ω resistor in the circuit shown in figure and verify answer by using Nodal analysis and Superposition theorem.  	**$I_{1\Omega} = 4A$**
Q-2	**K3**	**CO1**	Caculate the in 2 Ω by using mesh analysis and verify answer by using Nodal and Thevenin's theorem  	**$I_{2\Omega}$=0.731A**
Q-3	**K3**	**CO1**	Calculate the current in 10 Ω by using Mesh analysis, Nodal, analysis Superposition, Thevenin and Norton's Theorem.  	**$I_{10\Omega}$= 18A**

Q-4	**K3**	**CO1**	.Determine the Voltage $\mathbf{V_o}$ in the given Circuit by Mesh, Nodal and Superposition theorem. 	$\mathbf{V_o = 10\ V}$
Q-5	**K3**	**CO1**	Determine Current I by using Nodal analysis 	**I = 0.838 A**
Q-6	**K3**	**CO1**	Determine Voltage drop in 3Ω Resistor Using Mesh and Nodal Analysis. 	**V = 2.5 V**
Q-7	**K3**	**CO1**	Calculate the current in 20 Ω by using Mesh analysis, Nodal, analysis Superposition, Thevenin and Norton's Theorem. 	$\mathbf{I_{20\Omega} = 2A}$
Q-8	**K3**	**CO1**	Calculate the current in AB branch by using Mesh analysis, Nodal, analysis, Thevenin and Norton's Theorem 	$\mathbf{I_{AB} = 1A}$ B to A

Q-9	**K3**	**CO1**	Determine current in 6Ω resistance using source transformation. 	**$I_{6\Omega}$ = 0.79A**
Q-10	**K3**	**CO1**	Calculate the current in 6 Ω by using Mesh, Nodal analysis, Superposition, Thevenin and Norton's Theorem. 	**$I_{6\Omega}$ = 0.667 A**